

AKASH KARMAKAR

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Experience

AIMPAAS, Alternate Investment Management Platform as a Service, Software Consultant **Oct 2025 – Present**

- Developed an automated primary-backup failover and recovery framework integrating AWS EC2, ALB target groups, MariaDB, Redis and cron orchestration, enabling seamless role switching and near-zero downtime for production trading and market-data systems.
- Modernized market-data and analytics APIs for institutional clients across hedge funds and global banks.
- Designed and implemented an S3-backed state persistence and restoration pipeline to snapshot in-memory trading state, including shared-memory dumps every 5 minutes, cron configurations, binaries and logs.

FBN Securities, Developer Intern **Sep 2025 – Dec 2025**

- Architected and built a full-stack datapoint platform used by internal analysts to ingest long-form earnings-call video into transcripts (> 1 hour, 300MB), significantly reducing manual review time and accelerating research.
- Implemented a React write/edit interface with a Node.js/PostgreSQL backend enabling analysts to create, edit, annotate and structure unstructured commentary by company, sub-sector, sentiment, consultants and key themes into research-ready data points.

WorldQuant, Quantitative Research Consultant **Apr 2025 – Sep 2025**

- Formulated and simulated trading signals across US, EMEA and APAC markets, achieving exceptional performance metrics, including a portfolio Sharpe ratio of 2.24 and a RoMaD of 9.21 in simulation.
- Optimized market-neutral portfolios and created alphas delivering annualized returns exceeding 15% with low drawdowns, validated through multi-regime analysis, parameter sensitivity tests and robustness checks to ensure stability and minimize overfitting.

Goprac, Software Engineer **Jun 2023 – Apr 2024**

- Collaborated with the product team and led the integration of an AI-driven personalized feedback system using skills-based scoring, optimizing assessment workflows, reducing friction by removing manual review and improving efficiency for 36,000+ users.
- Engineered an automated interview question upload pipeline featuring auto-validation and cluster balancing, boosting upload speed by 80% and reducing upload errors by 99%, thereby eliminating manual rework.

Celebal Technologies, Data Scientist **Jul 2022 – Jun 2023**

- Diagnosed and refactored inefficient SQL queries and assessed advanced data mining techniques and conducted exploratory data analysis (EDA) on unstructured data, leading to significant insights for feature engineering.

Achievements

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- (National Finalist) Global rank **14** out of **76,867** in the International Quant Championship **2025**.
 - Secured 3rd place in Zicklin Graduate Case Competition **2025** & Ranked globally **1056** in Google Kickstart **2021**
 - Completed Akuna Options **101** and invite-only Akuna Options **201** courses.

Technical Skills

Languages/Databases/Systems: C++, Java, Python, JavaScript, Node.js, React, SQL, PostgreSQL, Redis, Linux
Cloud & Concepts: AWS (EC2, S3, ALB), Distributed Systems, Data Pipelines, Fault-Tolerant Architecture, Git

Education

Baruch College, MS in Finance-Fintech - New York, GPA: 3.6/4.0 **Aug 2024 – May 2026**
Jadavpur University, Bachelor in Engineering - Kolkata, India, GPA: 3.7/4.0 **Jul 2018 – Jul 2022**

Projects

Dynamic Sector Rotation with Machine Learning Github link

- Designed and validated a LightGBM-based sector rotation strategy integrating PCA and K-Means clustering for market regime detection and ETF allocation optimization, achieving a Sharpe ratio of 1.12 in-sample. Conducted robust backtesting across 2014–2025 market regimes, outperforming the Pairs Portfolio benchmark in out-of-sample returns (8.76% vs. 7.35%) while controlling drawdowns.

Portfolio Optimization and Efficient-Frontier Visualizer Github link

- Portfolio optimizer leveraging historical stock data to simulate 10,000 portfolios and visualize efficient frontier.